Development of magnetic liquid metal suspensions for magnetohydrodynamics

Florian Carle,[1] Kunlun Bai,[1] Joshua Casara,[2] Kyle Vanderlick,[3] and Eric Brown[1]

[1]*Department of Mechanical Engineering and Materials Science, Yale University, New Haven, CT 06511, USA*
[2]*School of Natural Sciences, University of California, Merced, CA 95343, USA*
[3]*Department of Chemical and Environmental Engineering, Yale University, New Haven, CT 06511, USA*

A new class of materials is developed that is a liquid with both high conductivity and magnetic susceptibility for magnetohydrodynamic (MHD) applications. We develop a general method for making such suspensions and demonstrate that various magnetic and non-magnetic metal particles, from 40 nm - 500 μm in diameter, can be suspended in liquid gallium and its alloys. The method uses an acid solution to prevent oxidation of the liquid metal and metallic particles, which allows wetting and thus suspending. We can increase the magnetic permeability by a factor of 5.0 by controlling the packing fraction of magnetic particles, which gives these materials the potential to exhibit strong MHD effects on the laboratory scale that are usually only observable in the cores of planets and stars. We can independently tune the viscosity by a factor of 230 by adding non-magnetic particles, which would allow independent control of MHD effects from turbulence.

Keywords: magnetohydrodynamics, dynamo, suspensions, magnetic susceptibility, liquid metal, eGaIn

arXiv:1512.02575v2 [cond-mat.mtrl-sci] 11 Dec 2015

INTRODUCTION

Magnetohydrodynamic (MHD) phenomena involve the interaction of magnetic fields with fluid flows. For example, many planets and stars exhibit a spontaneous dynamo effect in which a magnetic field is generated by flow of a conducting fluid. This, and many other MHD phenomena which depend on fluid flow advecting magnetic fields, occur at large magnetic Reynolds number Re_m, which is a characteristic ratio of advection to dissipation of electrical currents and magnetic fields in the flow. $Re_m = \sigma\mu_0(1+\chi)UL$, where U and L are characteristic velocity and length scales of the system, μ_0 is the permeability of free space, χ is the magnetic susceptibility of the fluid, and σ is its electrical conductivity. A high value of Re_m is easily achieved for the large scale L of planets and stars, but this is much harder to achieve on the smaller scale of a laboratory.

Our goal is to develop a material that can easily achieve a large χ and Re_m on the laboratory scale so that MHD phenomena may be more easily studied experimentally and used for practical applications. A spontaneous dynamo requires the highest $Re_m(\gtrsim 30)$ to achieve [1], so this Re_m is our ultimate target goal for materials design. To achieve such high Re_m, liquid metals and plasmas are traditionally used in experiments because they are the fluids with the highest conductivities σ. To contain plasmas requires specific magnetic field arrangements and a laminar flow, which is a challenge that makes their use limited in dynamo experiments [2]. Liquid sodium is the preferred working fluid for dynamo experiments [1, 3–5] because of its high conductivity ($\sigma = 9.6 \cdot 10^6$ S/m)[6], but can be a challenge to work with because it has a high melting point and explodes on contact with water. Even then, experiments have had to be at least 0.5 m in size and have taken at least 13 years to create a dynamo [1, 3, 4]. Gallium, which has lower conductivity ($\sigma = 3.4 \cdot 10^6$ S/m) than sodium, has been suitable for observing other MHD phenomena such as the magneto-rotational instability which only require $Re_m > 1$ [7]. A limitation of liquid metals is that at temperatures at which they are liquid, all metals have negligible $\chi \approx 10^{-6} - 10^{-5}$, which is the other material parameter that affects Re_m. We propose to improve on the achievable range of χ and Re_m by suspending magnetic particles into liquid metals to create a Magnetic Liquid Metal (MLM) with both a large χ from the particles and σ from the liquid metals.

Making MHD phenomena more accessible on the laboratory scale could also lead to development of new devices. For example, the high conductivity and magnetic susceptibility would be desirable in a MHD generator that converts thermal energy to electric energy without moving parts, which can harness the work done by a change in magnetization in an applied magnetic field, and move electric current generated by the dynamo effect through the conductor [8, 9]. The ability to rapidly switch on and off the magnetic response via a controlled magnetic field is also highly desired in devices. For example, in magnetorheological dampers, damping increases in response to a magnetic field due to both an increase in effective viscosity from the magnetorheological effect and eddy currents in the conductor [10, 11].

Suspension of magnetic particles in liquid metals has been attempted before. Attempts to suspend pure iron or nickel particles in non-oxidized gallium failed to make suspensions [12–14]. On the other hand, nickel [13, 15] and FeNbVB [14] particles coated with silica were suspended in gallium. However, in one of those cases the liquid metal was intentionally oxidized to allow suspending silica-coated particles [15]. In the other cases, we can infer that the liquid metals were oxidized based on the pictures reported which appear dull, rather than shiny like a pure liquid metal [13, 14]. Because oxidized liquid

metals have a thin oxidation film on their surface, they have a yield stress like a solid, and so do not flow as would be desired for MHD applications [16].

Martin *et al.* [17] suspended iron beads of diameter $d = 6.35$ mm into liquid gallium. These particles are larger than the capillary length ($\sqrt{\gamma/\rho g} = 3.2$ mm for gallium [18], where γ is the surface tension, ρ is the density and g the gravitational constant), which means that they are heavy enough that the stress due to their weight ($\sim \rho g d = 500$ Pa) is larger than the stress from surface tension ($\sim \gamma/d = 100$ Pa). The gravitational stress is also greater than the yield stress τ_y (≈ 100 Pa) due to the oxide layer [16]. Thus, particles can break through the surface to get into the bulk of the liquid metal regardless of surface tension or whether there is an oxidation layer. However, particles this large can break out through the surface just as easily, settle quickly under the effect of gravity, and are not expected to follow fluid flow due to their inertia, properties which may be undesirable in applications.

Magnetic particles have been successfully suspended into mercury, in which case it was implied that there was no oxidation [8, 19]. However, mercury is not a desirable liquid due to its toxicity. Furthermore, only very low packing fractions of magnetic particles were suspended in mercury; resulting in $\chi = 2 \cdot 10^{-4}$ for up to 2% by weight Fe-Ni-B particles [19] and $\chi = 3 \cdot 10^{-3}$ for a few percent iron particles coated with tin [8], too small to allow a significant increase in $Re_m \propto 1 + \chi$.

To address these weaknesses of previous attempts to suspend magnetic particles in liquid metals, we present in this paper a new suspension method that can be generally applied to suspend a variety of different metallic particles into different liquid metals. By removing and preventing oxidation of the metallic particles with a flux (i.e. an acid), we make the surface wettable by liquid metals [20], which allows them to suspend into the liquid bulk. The flux also prevents and removes any oxide layer from the liquid metal so it behaves mechanically like a simple liquid [16]. The suspending method has the advantage that we can suspend both non-Brownian and Brownian particles; the latter is desirable since particles do not settle or behave inertially. By increasing the packing fraction of magnetic particles suspended up to 40% by volume, we are able to tune χ and thus Re_m over a wide range. We demonstrate this suspension method using liquid gallium and two of its alloys. These liquids have the advantage that they are liquid at or near room temperature, have low vapor pressure (i.e. they are not toxic like mercury [21]), and do not explode on contact with water like sodium, making them relatively easy to work with compared to other liquid metals. However, they do have the disadvantage that they corrode some metals, such as aluminum and copper. We explain in detail in this article how this suspension process works generally for a variety of materials so it can be easily reproduced and extended in other laboratories.

Additionally, by adding a third phase of non-magnetic particles into the suspension, we can additionally tune the viscosity η of the fluid. In particular, this is useful for tuning the Reynolds number $Re = \rho U L/\eta$ (or, equivalently, the magnetic Prandtl number $Pr_m = Re_m/Re$) independently of Re_m. Since both Re and Re_m scale with U and L, independent control cannot be done over a significant range with liquid metals alone. Since MHD phenomena happen for high Re_m, they are also at high Re and thus turbulent. Controlling separately the MHD control parameter Re_m and the turbulence control parameter Re may allow understanding how each separately contributes to MHD phenomena.

RESULTS

Method to produce magnetic liquid metals

We used several liquid metals as base liquids for the creation of MLM. The main one used in this study is eGaIn which is an eutectic alloy of gallium and indium [22] at the weight ratio 77/23 (melting point $T_{melt} = 15°$C, density $\rho = 6250$ kg/m^3, surface tension $\gamma = 624$ mN/m [23], electrical conductivity $\sigma = 3.44{\cdot}10^6$ S/m). The two metals were mixed at 35°C, at which gallium is a liquid and indium is a solid. The mixture was covered with a solution of hydrochloric acid (HCl), which cleans off and prevents oxidation [16]. Pure gallium ($T_{melt} = 29.7°$C, $\rho = 6095$ kg/m^3, $\sigma = 3.7{\cdot}10^6$ S/m) and a eutectic alloy of gallium, indium and tin (weight ratio 68.5/21.5/10 respectively, T_{melt} = -19°C, $\rho = 6440$ kg/m^3, $\sigma = 3.46{\cdot}10^6$ S/m) have also been used as base liquids. The gallium, indium and the eutectic alloy of gallium, indium and tin were purchased from Gallium Source and each have 99.99% purity.

When liquid metals are exposed to air, a thin solid oxide layer forms at the surface that appears dull (Fig. 1g), while the inside remains pure non-oxidized liquid metal protected from air. This oxidized liquid metal wets non-metallic surfaces and does not wet to metals [16]. Indeed, we found that oxidized iron particles which were mixed with oxidized eGaIn did not suspend into the bulk and instead stuck to the oxide skin at the surface. This suggests the high interfacial tension is a barrier to suspending particles in liquid metals. For a liquid to wet a solid, a rule of thumb is that it should typically be less polarizable than the solid surface, so that the liquid-solid van der Waals attraction is stronger than the liquid-liquid attraction [20, 24]. Thus, we propose that using a flux to remove the less-conductive oxide layer ($\sigma = 10^3$ S/m) from the iron particles to make the surface more conductive ($\sigma = 1.04 \cdot 10^7$ S/m) and thus more polarizable could enable eGaIn ($\sigma = 3.44 \cdot 10^6$ S/m) to wet the particles (much like soldering) and allow them to suspend. We use

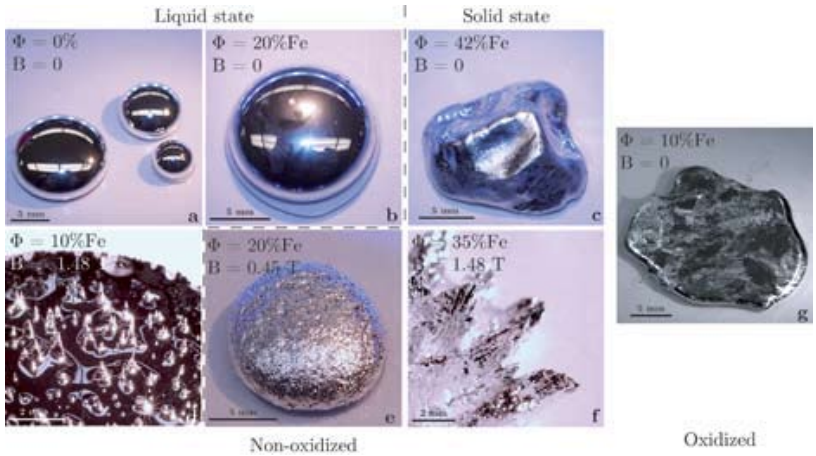

FIG. 1: Various phenomena observed in magnetic liquid metals (MLM). The suspension is 29 μm diameter iron particles in eGaIn, at values of packing fractions ϕ and applied constant magnetic fields B given in the panels. (a) eGaIn. (b) A suspension, which appeared similar to the pure eGaIn. (c) A yield stress appeared for ϕ above the liquid-solid transition ϕ_c. (d) The Rosensweig instability (e) The magnetorheological effect. (f) The combined effects of panels d and e. The dashed line between panels separates the liquid and solid states, also visible by the change of reflectivity: liquid metal states appear shiny because they are conductive and can take smooth shapes with minimal surface areas. Solid state are less shiny because the particles poke through the liquid-air interface, making the surface rough. They also have a yield stress that prevents them from flowing to obtain a minimal surface area. The samples in panels a-f were rinsed with HCl to remove oxidation shortly before taking the pictures, and small amounts of remaining HCl are visible in some of the pictures. (g) A sample exposed to air without HCL, which oxidized, causing it to appear less shiny, and develop a yield stress.

an acid solution as the flux. Additionally, the acid can remove the oxide skin of the liquid metal to eliminate the yield stress, allowing the liquid metal to flow like a simple liquid and form into a shape with a minimal surface area [16] (Fig. 1a and Movie 1 (SI)).

The MLM were made by mixing the liquid metal (typically 2mL) with varying packing fractions ϕ (by volume) of metallic particles and a hydrochloric acid (HCl) solution at room temperature. Enough HCl was used to completely cover the metals so it cleaned off any oxidation and prevented further contact between the metals and air (typically 10 mL HCL for 2 mL of liquid metal). The mixing process is shown in Movie 2 (SI) for iron particles ($\rho = 7874$ kg/m^3) with mean diameter 29 μm (the median 90% range for particle diameters is 18-40 μm, and there are particles as small as 1 μm and as large as 219 μm) suspended in eGaIn at $\phi = 10\%$ and a HCl solution with pH = 0.9. Particles that came into contact with the liquid metal absorbed and suspended into the bulk within a few seconds of contact. Contact could be achieved by initially placing the particles on top of the eGaIn before adding the HCl if the particles were big enough ($\gtrsim 1\mu$m) to settle under gravity. Brownian particles ($\lesssim 1\mu$m) instead suspended into the HCl solution (appearing gray), in which case contact between the particles and liquid metal could be achieved by vigorous shaking or stirring of the solution for 30 minutes (after which the HCl solution became transparent). If initially oxidized particles were mixed with oxidized eGaIn, the particles did not suspended to the bulk of the liquid metal until the HCL was added (Movie 3 (SI)). This demonstrates the need for the acid for suspending to occur.

The role of the liquid metal oxidation was tested by varying the pH of the HCL solution. For pH $< 0.95\pm0.05$, the surface of the eGaIn appeared shiny, indicating an absence of an oxide layer, and the particles mixed into the bulk of the eGaIn to form a suspension. If instead the HCl solution had pH $> 0.95 \pm 0.05$, the particles did not suspend into the bulk of the eGaIn even after sitting in contact with eGaIn in the acid bath for 24 hours or stirring the sample for 30 seconds. This critical pH value is consistent with Xu *et al.* [16], who observed that the oxide skin disappeared and wetting of eGaIn to metallic surfaces occurred only for pH $\leq 1\pm0.15$. Thus, at higher pH our eGaIn was oxidized, preventing it from wetting the particles. This confirms that wetting of the particles by the liquid metal is necessary for the particles to suspend into the liquid.

Movies 1 and 3 (SI) show that the oxidation removal from the liquid metal and particles respectively happen in matter of seconds, allowing the liquid metal to form a minimal surface and wet to non-oxidized metal surfaces [16]. At this point the particles started to suspend. A longer-lasting chemical reaction that creates bubbles and fumes also occurs. This same reaction was also found to occur between just HCL and iron particles, without eGaIn, meaning the eGaIn was not involved. If the iron particles were soaked in HCl for 16 hours before contact-

ing the eGaIn, they did not suspend anymore, presumably because the particle surfaces were converted to less conductive iron chloride ($\sigma = 10^{-9}$ S/m) from the chemical reaction with HCl. Beyond cleaning the oxidation in the first few seconds, the bubble-producing chemical reaction between iron and HCl did not aid the suspension process.

Robustness of the suspension process

To demonstrate the robustness of the suspension process for various liquid metal bases, the same 29 μm iron particles have also been successfully suspended using the same process into pure liquid gallium at 50°C (above its freezing point of 29.7°C) and into a eutectic alloy of gallium, indium and tin at room temperature. We also tried the same process to suspend various materials and sizes of particles (40 nm to 500 μm diameter) in eGaIn, shown in Table I. The data shows that particles with a higher conductivity σ than the liquid metal base (zinc, nickel, iron, and steel) were wetted and suspended into the bulk, while surfaces with lower σ surfaces (iron oxide, iron chloride, soda-lime glass, polystyrene, fused zirconia silica) were not wetted or suspended into the bulk. Of those, titanium, nichrome and stainless steel surfaces were found to not be wetted, but we did not obtain particles to test whether they suspended. These observations confirm the proposal that the particles should be more conductive than the liquid metal to be wetted [20], and this is a necessary and sufficient condition for suspending in our experiments.

Now that we understand a general procedure for suspending particles into liquid metals, we can explain why previous studies could not suspend uncoated metallic particles into liquid metal, but could suspend particles with non-metallic coatings into liquid metals [13–15]. The surfaces of the samples shown in those articles were not shiny, looking more like Fig. 1g, which is a clear indication of oxidation. The oxide layer prevented the liquid metal from wetting uncoated metal particles, but allowed it to wet nonmetals [16].

Qualitative properties

Non-oxidezed MLM (Fig. 1b) looked like pure liquid metal (Fig. 1a). The surface was shiny, and the lack of a yield stress allowed a droplet to flow freely if its container was tilted, or the droplet would take a shape with minimal surface area at rest due to the high surface tension of the liquid. In contrast, if the liquid metal surface was oxidized, it appeared less shiny (Fig. 1g).

When ϕ was increased above a critical packing fraction ϕ_c corresponding to the liquid-solid transition ($\phi_c = 40.7 \pm 0.5\%$ for 29 μm diameter iron particles in eGaIn),

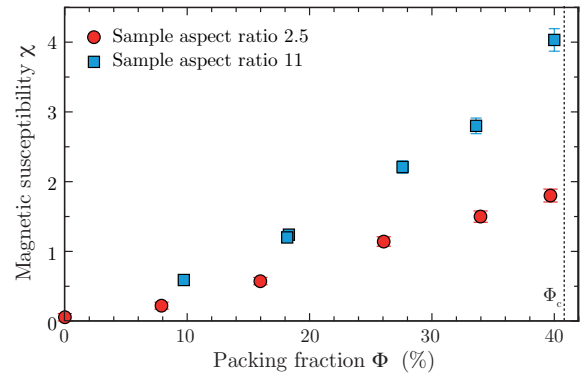

FIG. 2: Magnetic susceptibility χ of suspensions of 29 μm diameter iron particles in eGaIn as a function of packing fraction ϕ, for cylindrical samples of aspect ratio 2.5 (circles) and aspect ratio 11 (squares). The dashed line represents the liquid-solid phase transition. By tuning ϕ of magnetic particles, $Re_m \propto 1 + \chi$ increased by a factor up to 5.0 relative to pure eGaIn.

the particles became so densely packed that they poked through the liquid air-interface, making the surface appear rough or matte and less shiny, and resulting in a yield stress that allowed the sample to maintain shapes with non-minimal surface areas (Fig. 1c) [25].

We can compare the magnetic properties of MLM in constant magnetic fields B to other ferrofluids and magnetorheological fluids. For low ϕ and high B, MLM behaved as other ferrofluids; the surface deformed into sharp peaks that align with the magnetic field lines (Fig. 1d), a consequence of the Rosensweig instability [9]. For higher ϕ and lower B, the MLM exhibited the magnetorheological effect (Fig. 1e): a yield stress due to induced dipole-dipole interaction between magnetic particles [26, 27]. The transition from a simple liquid to these magnetic states in response to introducing or removing a magnetic field happens in a fraction of a second and is reversible (Movie 4 (SI)). All of these magnetic properties in MLM appear qualitatively similar to other ferrofluids and magnetorheological fluids in oils rather than liquid metals.

Accessible material parameter range

To identify the tunable range of magnetic susceptibility χ, we measured χ for different packing fractions ϕ of 29 μm diameter iron particles suspended in eGaIn. The measurements were taken using a gradiometer [28]. It measures an induced voltage due to magnetic induction in a solenoid in response to an applied alternating magnetic field. We calculate χ proportional to the increase in induced voltage when a sample is placed in the solenoid, corresponding to the effective material response to a magnetic field, which is modified from the intrinsic material

TABLE I: The suspension method has been demonstrated for a variety of magnetic and non-magnetic metallic particles suspended in eGaIn, listed in the table. The suspensions are ordered by decreasing electrical conductivity σ, showing the particle materials with higher σ than eGaIn ($\sigma = 3.44 \cdot 10^6$ S/m, horizontal line), were wetted and suspended, while particle materials with lower σ did not wet or suspend. The particles in bold were those used for quantitative measurements. References for material source and purity (%): [a] U.S. Nano (99.9), [b] Alfa Aesar (97.5), [c] Chemical Store (99), [d] Chemical Store (99.5), [e] Chemical Store (97.7), [f] Steve Spangler Science (99.5), [g] iron was oxidized to obtain an iron oxide surface, [h] iron was reacted with HCl to obtain an iron(III) chloride surface.

Material	Source (purity %)	Mean Diameter (μm)	Conductivity (S/m)	Wetted & Suspended
Zinc	(see caption)	0.100[a], **8**[b], 500[c]	$\mathbf{1.69 \cdot 10^7}$	✓
Nickel	Chemical Store (99)	175	$1.47 \cdot 10^7$	✓
Iron	(see caption)	0.040[a], 0.10[a], 0.90[a], 8[d], **29**[d], 45[e], 450[f]	$\mathbf{1.04 \cdot 10^7}$	✓
Steel	Chemical Store (99.4)	175	$1 \cdot 10^7$	✓
Stainless Steel	McMaster-Carr	Plate	$1.36 \cdot 10^6$	✗
Titanium	McMaster-Carr	Plate	$2.40 \cdot 10^6$	✗
Nichrome	VWR(99)	Wire	$6.67 \cdot 10^5$	✗
Iron Oxide[g]	Chemical Store (99.5)	29	10^3	✗
Iron(III) Chloride[h]	Chemical Store (99.5)	29	10^{-9}	✗
Soda Lime glass	Mo-Sci Corp.	125	10^{-13}	✗
Polystyrene	Sigma-Aldrich	200	10^{-15}	✗
fused Zirconia Silica	Mo-Sci Corp.	100	10^{-18}	✗

property by the demagnetization effect [29]. The resulting χ as function of ϕ is shown in Fig. 2 for two different sample aspect ratios, up to $\phi = 40\%$, which is the maximum we obtained before the liquid-solid transition ϕ_c. The nearly linear trend of increasing χ with packing fraction is expected due to the increase of magnetic material [30]. However, in detail, χ is expected to depend in a complicated way on ϕ, the aspect ratio of the samples, and the aspect ratio of the particles [29]. These dependencies will be reported in detail in a follow-up article [31]. For the nearly spherical particles shown in Fig. 2, we reached up to $\chi = 4.03 \pm 0.16$ for a sample aspect ratio of 11 and $\phi = 40\%$. This results in an increase of a factor up to 5.0 in $Re_m \propto 1 + \chi$, and a much bigger increases in χ from the pure liquid eGaIn ($\chi = 2.19 \cdot 10^{-6}$) [32] by a factor up to $1.8 \cdot 10^6$.

To tune the Reynolds number Re independently of the magnetic Reynolds number Re_m requires varying the fluid viscosity η without changing χ or σ significantly. We measured η using a rheometer setup and method similar to Xu *et al.* [16]. MLM suspensions we measured had a linear relationship between shear stress and shear rate – as a Newtonian or simple fluid does – over a range of shear rates from 1 to 1000/s. Fig. 3 shows η for a suspension of 29 μm diameter iron particles at various packing fractions ϕ_{Fe} in eGaIn as a baseline for comparison (circles). The trend of increasing η with ϕ_{Fe} is typical of suspensions, although the increase does not achieve the divergence at ϕ_c (= 40.5% for this suspension) of idealized theory [33], as in practice this divergence tends to be cut off. While η of the suspension of iron particles in eGaIn shows a significant increase on its own, the range of η independently tunable from χ can be obtained by adding non-magnetic

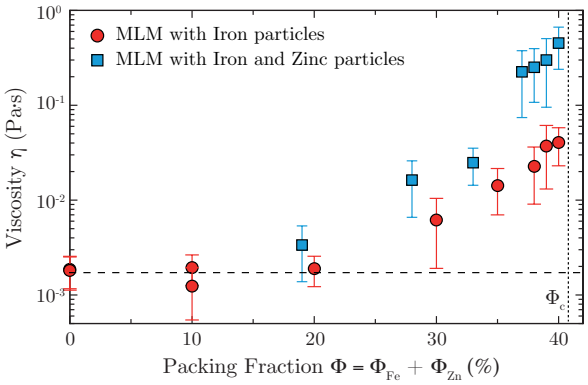

FIG. 3: Circles: Viscosity η of a suspension of 29 μm diameter iron particles in eGaIn with varying packing fraction ϕ_{Fe}. Squares: η for the same suspension with fixed $\phi_{Fe} = 10\%$ and additionally 8 μm diameter zinc particles at variable packing fraction ϕ_{Zn}. The vertical dashed line represents the liquid-solid transition ϕ_c. The horizontal dashed line represents η for pure eGaIn ($\phi = 0$). By tuning ϕ_{Zn} of non-magnetic zinc particles at fixed ϕ_{Fe}, η increased by a factor up to 230 without changing χ.

particles in addition to the magnetic particles already in the MLM suspension. We compare a sample with a fixed $\phi_{Fe} = 10\%$ of iron particles with a sample that additionally has 8 μm diameter non-magnetic zinc particles (nominally 6-9 μm, $\rho = 7140$ kg/m^3) suspended at packing fractions ϕ_{Zn} up to the liquid-solid transition ϕ_c. We plot η in Fig. 3 for this series (squares), as a function of total packing fraction $\phi = \phi_{Fe} + \phi_{Zn}$. For this series, η increased by a factor of 230 larger than the sample with $\phi_{Fe} = 10\%$ and $\phi_{Zn} = 0$ at fixed χ before ϕ_c was reached.

DISCUSSION AND CONCLUSIONS

In summary, we demonstrate how to make a novel material: magnetic liquid metal (MLM), with both high electrical conductivity σ and magnetic susceptibility χ, by suspending magnetic particles in liquid metal. The suspension process is general enough to apply to a wide range of particle materials and particle sizes from 40 nm to 500 μm (Table I), as well as three different liquid metal bases. The suspension method uses a HCl solution as a flux to eliminate oxidation from both the metal particles and liquid, which allows the liquid metal to wet the particles as long as the particles have higher conductivity σ. The suspension method should be easily reproducible in other laboratories as it does not require significant experience in any techniques, and the materials are easily purchased, relatively safe, and can be used at room temperature.

The magnetic susceptibility χ can be increased by suspending magnetic particles to different packing fractions ϕ. MHD effects relating to the advection or generation of magnetic fields by fluid flow depend on $Re_m \propto \sigma(1+\chi)$ which was increased by up to a factor of 5.6 compared to the pure liquid metal; $1+\chi$ increases by 5.0 (Fig. 2) and σ by 13% (we measured $\sigma = 3.9 \cdot 10^6$ S/m and $\chi = 4.0$ for $\phi = 40\%$ iron). This is enough to reach a 2.0 times higher Re_m in eGain-based MLM than sodium ($\sigma = 9.6 \cdot 10^6$ S/m at a temperature $T = 120°$C [6]) at the same values of U and L. Furthermore, if these suspending techniques can be applied to liquid sodium, the material properties achieved should allow MHD experiments to reach Re_m about 5 times higher than previous experiments, or alternatively shrinking the volume of the experiment by a factor of 5^3, which would significantly reduce challenge and cost. This would require a non-aqueous flux to prevent sodium oxidation.

The viscosity η can be increased independent of the magnetic susceptibility χ at a fixed packing fraction of magnetic particles by additionally suspending varying packing fractions of non-magnetic particles. By approaching the liquid-solid transition ϕ_c, we were able to increase η by factor of 230 (Fig. 3). The parameter space in terms of Re_m and Re that could be achieved based on the range of χ, σ, η, and ρ we measured is shown in Fig. 4. A modest experiment scale of $L = 0.1$ m and $U \leq 15$ m/s is large enough to reach $Re_m = 30$, the threshold for a spontaneous dynamo [1]. While fixed material properties (i.e. χ, σ,η, and ρ) only allow movement along a diagonal line of slope 1 in the parameter space by varying U or L, the ability to independently control the material properties, either by suspending particles (MLM) or changing temperature (plasma) allows access to both dimensions of the parameter space. The material properties of the fluid in MHD are typically characterized by the magnetic Prandtl number ($Pr_m = Re_m/Re$) which is fixed

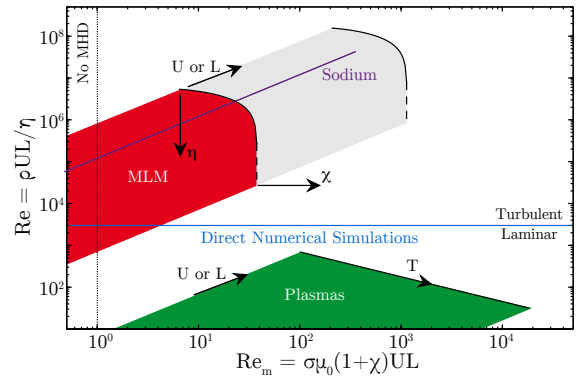

FIG. 4: Parameter range potentially accessible in terms of magnetic Reynolds number Re_m and Reynolds number Re due to the tunable material properties of MLM for a system with $U \leq 15$ m/s and $L = 0.1$ m (red region), or $L = 3$ m (gray region, corresponding to the size of the Maryland sodium facility [5] and the Wisconsin plasma facility [**?**]). The solid black line bounding the red regime uses fitted values of χ, σ, η, as function of the packing fraction ϕ measured for suspensions of iron particles in eGaIn from Figs 2 and 3 (we also measured an increase of σ by 13% with the addition of 40% iron particles in eGaIn, not shown), and ρ is calculated based on densities and volume fractions of the components. The vertical dashed line indicates the further decrease in Reynolds number due to increase in viscosity η by a factor of 230 over that of pure eGaIn with the addition of zinc particles (the same ratio achieved in Fig. 3), and assuming a fixed $\chi = 4.0$, $\rho = 6900$ kg/m^3 and $\sigma = 3.9 \cdot 10^6$ S/m (corresponding to $\phi = 40\%$ iron in eGaIn). The diagonal boundaries of the MLM regime correspond to varying U while other parameters are held constant. Purple line: liquid sodium at 120°C, for $U \leq 15$ m/s and $L = 3$ m used in the Maryland experiments [5]. Green region: parameter range proposed for helium plasmas with temperature $T = 1.0 \cdot 10^4 - 5.6 \cdot 10^5$°C, $U \leq 10$ km/s and $L = 3$ m as in the Wisconsin facility [**?**]. Direct numerical simulations and plasmas are confined to the laminar regime (bleu line, $Re \lesssim 3000$). MLM allow exploring both dimensions of this parameter regime independently, covering both turbulent and laminar flows, which cannot be achieved by other approaches. The arrows represent the direction on the parameter space achievable tuning the corresponding property.

for pure liquids. With MLM, we achieved a range from $Pr_m = 1.6 \cdot 10^{-6}$ for pure eGaIn, up to $Pr_m = 4.6 \cdot 10^{-4}$ for eGaIn with 10% iron and 30% zinc particles. An important result in Fig. 4 is that Re could span from the turbulent to the laminar regimes as a result of the ability to increase η. For example, $Re = 680$ could be reached while still achieving $Re_m = 1$ (the minimum threshold for most MHD phenomena). This would be the first approach to MHD that could span between the turbulent and laminar regimes and connect the parameter regimes of the different approaches; liquid metal experiments are always turbulent, and plasmas and direct numerical simulations are confined to the laminar regime due to computational cost and flow instabilities, respectively.

The other aspect of MHD is the Lorentz force on fluid flow due to magnetic fields, which also increases with χ. One source is dynamo effect which creates an induced magnetic field which applies a Lorentz force on the induced current. While this feedback mechanism exists in astrophysical MHD, no laboratory experiment has reached this regime yet. This regime could be achieved if the ratio of Lorentz stress to inertial stress $N^* = 2\sigma(1+\chi)^2 H^2 L/f\rho U \gtrsim 1$. If we assume $L = 0.1$ m, $f \approx 0.02$ as a typical value for a friction factor [34], and an induced magnetic field of $H = 3 \cdot 10^3$ A/m (the value obtained in the Cadarache dynamo at the threshold $Re_m = 30$ [1]), we can adjust U to find the value of N^* at which $Re_m = 30$ can still be obtained for different materials. We obtain $N^* = 0.07$ for liquid sodium ($\sigma = 9.6 \cdot 10^6$ S/m, $\rho = 927$ kg/m^3), $N^* = 0.2$ for our suspensions of $\phi = 40\%$ iron in eGaIn ($\chi = 4.0$, $\sigma = 3.9 \cdot 10^6$ S/m, $\rho = 6900$ kg/m^3), and $N^* = 2$ for a hypothetical suspension of liquid sodium with $\phi = 40\%$ iron particles ($\chi = 4$, $\sigma = 9.6 \cdot 10^6$ S/m, $\rho = 3700$ kg/m^3). This suggests that MLM with eGaIn could have Lorentz forces with a noticeable effect on the flow, while Lorentz forces could potentially be dominant for MLM in sodium.

The tunable range of $1+\chi$ and viscosity η for the materials reported are expected to be typical of MLM. The independently tunable range of η is expected to be insensitive to material, rather it depends on how far ϕ is from ϕ_c [25, 33]. There is potential that η could be increased beyond the factor of 230 we obtained; for example, a 3.5 decade increase of η with ϕ has been achieved with non-MLM suspensions while remaining Newtonian [35]. The increase in χ with magnetic particles is also predicted to be insensitive to the material as long as it is ferromagnetic [29], so we expect increases in Re_m of order 1-10 for nearly spherical particles. It also remains to be seen if χ will retain the higher values of large sample aspect ratios (Fig. 2) or achieve the even larger values predicted for larger particle and sample aspect ratios [29] in a turbulent flow where the magnetic field is not uniform or aligned with the sample or particles. If so, extreme sample aspect ratios could be taken advantage of in specific experiments to achieve much higher χ.

ACKNOWLEDGEMENTS

The authors acknowledge financial support from NSF Grant No. CBET-1255541 and AFOSR FA 9550-14-1-0337.

SUPPORTING INFORMATION

Movies available here: eng.yale.edu/brown/publications.html

Movie 1 - Oxidation removal

Two samples of eGaIn with packing fraction $\phi = 10\%$ iron (mean diameter of 29 μm) feature oxidation skin as they have been exposed to air for half an hour (slight oxidation) and three hours (strong oxidation). The oxide skin wets the plastic substrate preventing the sample to form into a shape with a minimal surface area. The HCl solution removes the oxide skin to eliminate the yield stress, allowing the liquid metal to flow like a simple liquid.

Movie 2 - Iron suspension

Oxidized iron particles (mean diameter of 29 μm) are poured in a HCL bath containing eGaIn. Particles that came into contact with the liquid metal absorbed and suspended into the bulk within a few seconds of contact. Particles too small to settle under gravity suspend into the HCl solution (appearing gray) and requires stirring to contact with the liquid metal.

Movie 3 - Oxidized particles

Dry particles are placed on top of eGaIn. They do not suspended to the bulk of the liquid metal despite stirring until HCL is added, demonstrating the need for the acid for suspending to occur.

Movie 4 - Interaction with a magnet

MLM are influenced by magnetic field. For a packing fraction of 10% of iron in eGaIn, the sample attracts to the magnets but the sample remains liquid. For a packing fraction of 20%, the sample solidifies. The transition from a simple liquid to these magnetic states in response to introducing or removing a magnetic field happens in a fraction of a second and is reversible.

[1] R. Monchaux, M. Berhanu, M. Bourgoin, M. Moulin, P. Odier, J.-F. Pinton, R. Volk, S. Fauve, N. Mordant, F. Pétrélis, A. Chiffaudel, F. Daviaud, B. Dubrulle, C. Gasquet, L. Marié, and F. Ravelet, "Generation of a Magnetic Field by Dynamo Action in a Turbulent Flow of Liquid Sodium," *Physical Review Letters*, vol. 98, no. 4, p. 044502, 2007.

[2] I. V. Khalzov, B. P. Brown, E. J. Kaplan, N. Katz, C. Paz-Soldan, K. Rahbarnia, E. J. Spence, and C. B. Forest, "Resistive and ferritic-wall plasma dynamos in a sphere," *Physics of Plasmas*, vol. 19, no. 10, p. 104501, 2012.

[3] R. Stieglitz and U. Müller, "Experimental demonstration of a homogeneous two-scale dynamo," *Physics of Fluids*, vol. 13, no. 3, pp. 561–564, 2001.

[4] A. Gailitis, O. Lielausis, S. Dement'ev, E. Platacis, A. Cifersons, G. Gerbeth, T. Gundrum, F. Stefani, M. Christen, H. Hänel, and G. Will, "Detection of a Flow Induced Magnetic Field Eigenmode in the Riga Dynamo Facility," *Physical Review Letters*, vol. 84, no. 19, pp. 4365–4368, 2000.

[5] D. S. Zimmerman, S. A. Triana, H.-C. Nataf, and D. P. Lathrop, "A turbulent, high magnetic Reynolds number experimental model of Earth's core," *Journal of Geophysical Research: Solid Earth*, vol. 119, no. 6, pp. 4538–4557, 2014.

[6] M. Berhanu, G. Verhille, J. Boisson, B. Gallet, C. Gissinger, S. Fauve, N. Mordant, F. Pétrélis, M. Bourgoin, P. Odier, J.-F. Pinton, N. Plihon, S. Aumaître, A. Chiffaudel, F. Daviaud, B. Dubrulle, and C. Pirat, "Dynamo regimes and transitions in the VKS experiment," *The European Physical Journal B*, vol. 77, no. 4, pp. 459–468, 2010.

[7] G. Rüdiger, R. Hollerbach, M. Schultz, and D. A. Shalybkov, "The stability of MHD Taylor-Couette flow with current-free spiral magnetic fields between conducting cylinders," *Astronomische Nachrichten*, vol. 326, no. 6, pp. 409–413, 2005.

[8] J. Popplewell, S. W. Charles, and R. Chantrell, "The long term stability of magnetic liquids for energy conversion devices," *Energy Conversion*, vol. 16, no. 3, pp. 133–138, 1977.

[9] R. E. Rosensweig, *Ferrohydrodynamics*. Mineola, New York: Dover Publications, 2014.

[10] M. Ripper and V. Endean, "Eddy-current braking-torque measurements on a thick copper disc," *Proceedings of the Institution of Electrical Engineers*, vol. 122, no. 3, pp. 301–302, 1975.

[11] J. Carlson, J. Holzheimer, D. Leroy, R. Marjoram, and D. Prindle, "Magnetorheological fluid brake," 1998. WO Patent App. PCT/US1997/008,853.

[12] R. Ito, G. Dodbiba, and T. Fujita, "Mr fluid of liquid gallium dispersing magnetic particles," *International Journal of Modern Physics B*, vol. 19, no. 07n09, pp. 1430–1436, 2005.

[13] L. F. Cao, H. S. Park, G. Dodbiba, and T. Fujita, "Dispersion of submicron Ni particles into liquid gallium," *Magnetohydrodynamics*, vol. 44, pp. 97–104, 2008.

[14] G. Dodbiba, K. Ono, H. S. Park, S. Matsuo, and T. Fujita, "FeNbVB alloy particles suspended in liquid gallium: investigating the magnetic properties of the MR suspension," *International Journal of Modern Physics B*, vol. 25, no. 07, pp. 947–955, 2011.

[15] M. Xiong, Y. Gao, and J. Liu, "Fabrication of magnetic nano liquid metal fluid through loading of Ni nanoparticles into gallium or its alloy," *Journal of Magnetism and Magnetic Materials*, vol. 354, pp. 279–283, 2014.

[16] Q. Xu, N. Oudalov, Q. Guo, H. M. Jaeger, and E. Brown, "Effect of oxidation on the mechanical properties of liquid gallium and eutectic gallium-indium," *Physics of Fluids*, vol. 24, no. 6, p. 063101, 2012.

[17] A. Martin, P. Odier, J.-F. Pinton, and S. Fauve, "Magnetic permeability of a diphasic flow, made of liquid gallium and iron beads," *The European Physical Journal B - Condensed Matter and Complex Systems*, vol. 18, no. 2, pp. 337–341, 2000.

[18] S. C. Hardy, "The surface tension of liquid gallium," *Journal of Crystal Growth*, vol. 71, no. 3, 1985.

[19] S. Linderoth, L. H. Rasmussen, and S. Mo/rup, "New methods for preparing mercury-based ferrofluids," *Journal of Applied Physics*, vol. 69, no. 8, pp. 5124–5126, 1991.

[20] P.-G. de Gennes, F. Brochard-Wyart, and D. Quéré, *Capillarity and Wetting Phenomena*. New York, NY: Springer New York, 2004.

[21] W. M. Haynes, ed., *CRC Handbook of Chemistry and Physics, 95th Edition*. CRC Press, 95 edition ed., 2014.

[22] L. De Boisbaudran, "Alloys of Indium and Gallium," *Journal of the Chemical Society (Great Britain)*, vol. 48, p. 700, 1885.

[23] M. D. Dickey, R. C. Chiechi, R. J. Larsen, E. A. Weiss, D. A. Weitz, and G. M. Whitesides, "Eutectic Gallium-Indium (EGaIn): A Liquid Metal Alloy for the Formation of Stable Structures in Microchannels at Room Temperature," *Adv. Funct. Mater.*, vol. 18, no. 7, pp. 1097–1104, 2008.

[24] H. W. Fox and W. A. Zisman, "The spreading of liquids on low-energy surfaces. III. Hydrocarbon surfaces," *Journal of Colloid Science*, vol. 7, no. 4, pp. 428–442, 1952.

[25] E. Brown, H. Zhang, N. A. Forman, B. W. Maynor, D. E. Betts, J. M. DeSimone, and H. M. Jaeger, "Shear thickening and jamming in densely packed suspensions of different particle shapes," *Physical Review E*, vol. 84, no. 3, p. 031408, 2011.

[26] J. Rabinow, "The Magnetic Fluid Clutch," *American Institute of Electrical Engineers, Transactions of the*, vol. 67, no. 2, pp. 1308–1315, 1948.

[27] C. F. Zukoski, "Material properties and the electrorheological response," *Annual Review of Materials Science*, vol. 23, no. 1, pp. 45–78, 1993.

[28] S. Tumanski, *Handbook of Magnetic Measurements*. CRC Press, 1 edition ed., 2011.

[29] R. Skomski, G. Hadjipanayis, and D. Sellmyer, "Effective Demagnetizing Factors of Complicated Particle Mixtures," *IEEE Transactions on Magnetics*, vol. 43, no. 6, pp. 2956–2958, 2007.

[30] P. Frick, S. Khripchenko, S. Denisov, D. Sokoloff, and J.-F. Pinton, "Effective magnetic permeability of a turbulent fluid with macroferroparticles," *The European Physical Journal B*, vol. 25, no. 4, pp. 399–402, 2002.

[31] K. Bai, A. Wahl, F. Carle, J. Casara, and E. Brown, "Effective susceptibility of magnetic suspensions," *in preparation*, 2016.

[32] B. P. Pashaey and V. V. Seleznev, "Magnetic susceptibility of gallium-indium alloys in liquid state," *Soviet Physics Journal*, vol. 16, no. 4, pp. 565–566, 1973.

[33] I. M. Krieger and T. J. Dougherty, "A Mechanism for Non-Newtonian Flow in Suspensions of Rigid Spheres," *Transactions of The Society of Rheology (1957-1977)*, vol. 3, no. 1, pp. 137–152, 1959.

[34] H. Schlichting and K. Gersten, *Boundary-Layer Theory*. Springer Science & Business Media, 2003.

[35] Z. Cheng, J. Zhu, P. M. Chaikin, S.-E. Phan, and W. B. Russel, "Nature of the divergence in low shear viscosity of colloidal hard-sphere dispersions," *Physical Review E*, vol. 65, no. 4, p. 041405, 2002.